Improving Prohibited-Item Detection and Instance Segmentation in X-ray Baggage Images through Background-Confusion Analysis and Occlusion-Aware Evaluation

1. Problem Definition and Research Gap

Automatic inspection of X-ray baggage images must locate prohibited items and delineate their visible regions despite weak texture, low contrast, semi-transparent overlap, clutter, large scale variation, and deliberate concealment. These conditions produce two safety-relevant error families: *missed threats*, particularly small or hidden items, and *false alarms*, where ordinary baggage structures resemble prohibited objects. A detector may improve overall average precision while leaving either problem unresolved; a correct box also does not guarantee an accurate instance mask.

Existing work addresses parts of this problem. The PIDray journal model uses normal baggage images, an image-level confidence node, a densely attentive feature pyramid, and Cascade Mask R-CNN. The recent occlusion-aware bilayer method uses SAM-based features and an explicitly supervised occluder decoder to improve masks under object-to-object overlap. However, background confusion, hidden-item proposal failure, and occlusion-boundary error are different mechanisms. The study therefore requires a controlled diagnosis before choosing an architectural change, and it must compare models only when dataset, supervision, split, and evaluator are aligned.

2. Aim, Proposed Direction, and Scope

The aim is to develop and evaluate a targeted improvement for prohibited-item detection and instance segmentation on realistic X-ray baggage images. The primary direction is **baseline-guided background-confusion reduction**: reproduce a strong box-and-mask baseline, quantify its failure modes, and implement one background-confusion-aware component only if the evidence shows that background errors are a substantial limitation.

The provisional solution family is a **feature- or proposal-level background-confusion refinement component** trained with normal images and hard false-positive regions. This is intentionally different from simply repeating PIDray's image-level R_0 gate: it would learn to suppress prohibited-item-like clutter within the detector while preserving proposals for difficult positives. The exact insertion point—feature pyramid, RPN/objectness, or RoI classifier—will be frozen after validation diagnostics. If the dominant error is instead proposal recall or mask leakage, the component will be redirected to that verified failure and the change will be stated before final testing.

The main study covers detection boxes, instance masks, background false alarms, difficult-case recall, and computational cost. Real-time performance is a secondary consideration rather than the central contribution. One- or two-stage efficient models may be reported as references only after the main baselines are stable.

3. Objectives and Research Questions

Item	Objective / research question	Required evidence
O1 / RQ1	Establish reproducible box-and-mask baselines and identify the errors that dominate on original PIDray.	Released-checkpoint test, reproduced run, Easy/Hard/Hidden and per-class metrics, proposal and error analysis.
O2 / RQ2	Determine how much error is attributable to background confusion rather than concealment, localization, classification, or mask boundaries.	Negative-image false alarms, matched/unmatched prediction analysis, RPN recall, and qualitative audit.
O3 / RQ3	Design one targeted component from the observed failure and test whether it improves the unchanged baseline.	Frozen hypothesis, controlled ablation, same data/supervision/evaluator, and validation-selected settings.
O4 / RQ4	Determine whether any gain is reliable, useful on difficult cases, and computationally acceptable.	Repeated principal runs where feasible, uncertainty, subset results, latency, memory, and parameter count.

4. Datasets and Experimental Tracks

Dataset	Role	Protocol
PIDray	Required primary track	Train and validate from the official training portion. Retain both positive and normal images. Evaluate the frozen model separately on Easy, Hard, and Hidden using boxes and masks.
PIDray-A	Conditional occlusion track	Use only for methods requiring the additional occlusion-area masks, principally Bilayer + SAM. Report separately from original PIDray because supervision differs.
PIXray / PIXray-A	Optional generalization track	Run after the primary study. State the 15-class taxonomy, split, annotation version, and whether ordinary or occlusion-augmented supervision is used.

Split rule. Easy, Hard, and Hidden are test subsets. They will not be used for validation, threshold selection, component design, hard-negative mining, augmentation selection, or checkpoint choice. A fixed development validation partition will be sampled only from the official PIDray training portion, stratified by positive/negative status and class where feasible.

5. Baselines and Comparison Roles

Model	Correct role	Use in this study
Cascade Mask R-CNN	Common implementation anchor	First reproducible box-and-mask model; supports proposal, box, and mask diagnostics.
SDANet	Earlier PIDray comparison	Keep as a separately named published method; do not use its name for the full 2023 journal system.
PIDray journal baseline / PIDrayNet	Primary benchmark-aligned baseline	Full divide-and-conquer Cascade Mask R-CNN with R_0 screening and AttFPN; reproduce on original PIDray.
Occlusion-Aware Bilayer + SAM	Strong specialized baseline	Reproduce on PIDray-A/PIXray-A only when extra annotations and compute are available; otherwise cite as an external result.
YOLO11-seg / RT-DETRv2	Optional efficiency references	Add only if required by the final research question; RT-DETRv2 is detection-only unless paired with a separate mask mechanism.

6. Step-by-Step Methodology

1. **Freeze the study contract.** Finalize the title, problem statement, primary outcome, baseline identities, data tracks, and rules separating published, reproduced, and proposed-method values.
2. **Audit data and annotations.** Record release identifiers, image and annotation checksums, class order, positive/negative counts, split membership, mask validity, and image-level difficulty labels. Render random samples, including empty annotations and overlapping instances.
3. **Validate the evaluator.** Test COCO box/mask evaluation with controlled predictions. Fix confidence handling, IoU conventions, subset filters, and the hardware protocol before comparing models.
4. **Evaluate released weights.** Run each official checkpoint before retraining. This isolates data-loader and evaluator problems from optimization differences.
5. **Reproduce the common anchor and PIDray baseline.** Begin with a one-batch test and a 1–5% smoke run, then train Cascade Mask R-CNN and the full PIDray journal model. Preserve negative images and archive the exact environment and configuration.
6. **Construct a validation error profile.** Separate background false positives, missed instances, proposal failures, class errors, localization errors, duplicates, mask leakage/fragmentation, small objects, and hidden items. Rank each category by count, rate, confidence, class, and operational importance.
7. **Select the intervention.** Write one causal hypothesis. For example: "hard background regions are assigned excessive RPN/RoI objectness; adding background-aware refinement will lower false positives per negative image without reducing Hidden recall." Freeze the module, insertion point, loss, and primary ablation before full training.
8. **Run controlled ablations.** Compare unchanged baseline, each indispensable component, and the complete method under the same data, supervision, initialization, schedule, evaluator, and tuning budget. Repeat the principal comparison with multiple seeds where feasible.
9. **Freeze and test.** Choose the checkpoint and operating threshold on validation data, then evaluate once on Easy, Hard, and Hidden. Run the occlusion or generalization track only after the primary result is complete.

7. Planned Experiment Matrix

Run	Model	Data / supervision	Question
E0	Released PIDray checkpoint	Original PIDray	Is the data/evaluator integration correct?
E1	Cascade Mask R-CNN	Original PIDray boxes and masks	What is the common anchor performance and error profile?
E2	PIDray journal baseline	E1 supervision plus normal images	What do R_0 and AttFPN contribute under the official design?
E3	E1 or E2 + proposed component	Exactly the matched E1/E2 protocol	Does the failure-driven component improve the prespecified outcome?
E4	Component ablations	Same as E3	Which part causes the improvement and at what cost?
E5	Bilayer + SAM	PIDray-A with occlusion masks	Does explicit occlusion supervision improve mask quality under overlap?
E6	Baseline vs. proposed method	Optional PIXray track	Does the effect generalize under a separately documented protocol?

8. Evaluation and Decision Rules

Dimension	Measures	Decision rule
Detection	Box AP, AP50, AP75, per-class and subset AP; proposal recall	Report Easy/Hard/Hidden separately; distinguish proposal failure from later-stage error.
Segmentation	Mask AP, AP50, AP75; matched-mask IoU/Dice and boundary measure	Do not infer mask improvement from box AP alone.
Background confusion	FP per negative image, negative-image alarm rate, high-confidence FP count, precision–recall	A useful gain lowers background errors without materially reducing difficult-positive recall.
Reliability	Recall, precision, F1; mean/SD and confidence interval where feasible	Select thresholds on validation only; principal conclusions should be directionally consistent across runs.
Efficiency	Parameters, peak GPU memory, training time, latency, throughput	Fix GPU, image size, batch, precision, warm-up, preprocessing, and measurement repetitions.

The proposed method is accepted only if it improves the prespecified primary metric, addresses the diagnosed error, remains useful on the relevant difficult subset, and reports its computational cost. A method that merely shifts errors—for example, fewer false alarms but substantially more hidden-item misses—will not be claimed as an overall improvement.

9. Expected Contribution and Immediate Order

The expected contribution is not a broad collection of modules. It is a reproducible, evidence-driven improvement supported by: (i) a verified PIDray baseline; (ii) a quantitative taxonomy connecting background confusion, hidden-object recall, and mask error; (iii) one justified component with controlled ablations; and (iv) a transparent accuracy–reliability–efficiency evaluation.

Implementation should begin in this order: data manifest and train-derived validation split; evaluator tests; released-checkpoint evaluation; Cascade Mask R-CNN smoke/full run; PIDray journal reproduction; validation error analysis; component selection; ablations and final test; then the PIDray-A/PIXray track if feasible. Every result row must be traceable to a code commit, configuration, dataset manifest, seed, checkpoint, prediction file, evaluator log, and hardware record.

Primary Sources

L. Zhang et al., "PIDray: A Large-Scale X-ray Benchmark for Real-World Prohibited Item Detection," *IJCV*, 2023. `https://arxiv.org/abs/2211.10763`. Official code: `https://github.com/lutao2021/PIDray`.
Y. Ren et al., "Prohibited Items Segmentation via Occlusion-Aware Bilayer Modeling," *ICME*, 2025. `https://arxiv.org/abs/2506.11661`. Official code: `https://github.com/Ryh1218/Occ`.